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EVALUATION AND COMPARISON OF TWO EFFICIENT PROBABILISTIC PRIMALITY TESTING ALGORITHMS

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Abstract. We analyse two recent probabilistic primality testing algorithms; the first one is derived from Miller [6] in a formulation given by Rabin [7], and the second one is from Solovay and Strassen [9]. Both decide whether or not an odd number n is prime in time $O(m, \log n\, M(n))$ with an error probability less than α^m, for some $0 \leq \alpha < \frac{1}{2}$. Our comparison shows that the first algorithm is always more efficient than the second, both in probabilistic and algorithmic terms.

1. Introduction

For every odd integer n we define a primality test by means of a predicate $T(n, a)$, where $1 \leq a < n$, with the following properties:

(1) the number n is composite iff $T(n, a)$ holds for some a,

(2) the function $T(n, a)$ can be efficiently computed,

(3) when n is composite, let $L(n)$ be the set of integers a for which $T(n, a)$ does not hold; then $|L(n)|/(n-1) \leq \frac{1}{2}$.

Such a test allows us to build an efficient probabilistic primality testing algorithm as follows: we produce a sequence of m independant random integers $\{a_i\}$, with $1 \leq a_i \leq n$. If one of these numbers satisfies the test, then n is composite, otherwise we declare n to be prime with a probability no greater than 2^{-m} of making the wrong choice.

For a given test, we note $\alpha_n = |L(n)|/(n-1)$ the probability that the composite number n is declared prime. The error probability after m iterations of the test is $(\alpha_n)^m$. Calling t the computation time of one test, it is easy to see that the ratio between the cost of the whole algorithm and the logarithm of its error probability

$$\mu_n = \frac{mt}{-\log(\alpha_n^m)} = \frac{t}{-\log \alpha_n},$$

does not depend on number m of iterations performed. Solovay and Strassen [9] have thus proposed to compare probabilistic tests by means of the measure μ_n.

Through an analysis of the error probabilities and the cost of both tests, we demonstrate that the Miller–Rabin test is more efficient than Solovay–Strassen's, according to this (and any) measure.

2. Notations and Fundamental Lemmas

Throughout this paper the prime decomposition of n is $n = p_1^{a_1} \cdots p_r^{a_r}$. The greatest common divisor of a and b is noted (a, b). The group of $\mathbb{Z}_n$'s units is $\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n;\ (a, n) = 1\}$. The function v_2 is the dyadic valuation; thus $a = 2^{v_2(a)}a'$, where a' is odd. For $n = p_1^{a_1} \cdots p_r^{a_r}$, we note $\nu_0 = v_2(n-1)$, $\nu_i = \nu_2(p_i - 1)$ and $\nu = \inf_{1 \le i \le r}\{\nu_i\}$; then we have $n - 1 = 2^{\nu_0}n'$ and $p_i - 1 = 2^{\nu_i}p_i'$ for any i. For any odd integer n, the Jacobi symbol is denoted $(\frac{a}{n})$.

The Euler totient function $\phi(n) = \prod_{1 \le i \le r} p_i^{a_i - 1}\ (p_i - 1)$, the Carmichaël functions $\lambda(n) = \mathrm{lcm}_{1 \le i \le r}\{\phi(p_i)\}$ and $\lambda'(n) = \mathrm{lcm}_{1 \le i \le r}\{p_i - 1\}$ receive their usual notation.

By $M(n)$ we denote the cost of multiplying two integers $\le n$; it is known by Schönhage–Strassen algorithm [8] that $M(n) = O(\log n \operatorname{llog} n \operatorname{lllog} n)$.

When n has a primitive root, and g is a generator of $\mathbb{Z}_n^*$, we let $\mathrm{ind}_g a$ be the index of a relatively to g; thus $g^{\mathrm{ind}_g a} \equiv a[n]$ for any a in $\mathbb{Z}_n^*$.

The next sections use, for enumerations, the number of solutions to the equation

$$x^k \equiv y[n]$$

in the unknown x; for our purposes, the right member y is equal to 1 or -1, but it is no harder to solve the general congruence:

Lemma 1. *Let $n = p_1^{a_1} \cdots p_r^{a_r}$ be an odd integer, y relatively prime to n and k a positive integer; the congruence $x^k \equiv y[n]$ has a solution if and only if $(k, \phi(p_i^{a_i}))$ divides $\mathrm{ind}_{g_i}(y)$ for all $1 \le i \le r$; if it has a solution, then there are exactly*

$$\prod_{1 \le i \le r} (k, \phi(p_i^{a_i}))$$

solutions modulo n.

Proof. Taking modulos, the congruence $x^k \equiv y[n]$ is equivalent to $x^k \equiv y[p_i^{a_i}]$, for any prime divisor p_i. Using indexes, the last congruence becomes $k\,\mathrm{ind}_{g_i}(x) \equiv \mathrm{ind}_{g_i}(y)[\phi(p_i^{a_i})]$, which has a solution if and only if $d_i = (k, \phi(p_i^{a_i}))$ divides $\mathrm{ind}_{g_i}(y)$. For a given solution x_0 we rewrite the congruence:

$$k(\mathrm{ind}_{g_i}(xx_0^{-1})) \equiv 0[\phi(p_i^{a_i})], \text{ and it comes}$$

$$\mathrm{ind}_{g_i}(xx_0^{-1}) = t_i \frac{\phi(p_i^{a_i})}{d_i} \quad \text{where } 0 \le t_i < d_i;$$

thus the congruence has either 0 or d_i solutions, and when grouping we complete the proof.

We must remark that the number of solutions is independant on the value of y. The relevant cases for us are

$y=1$: the congruence always has solutions.

$y=-1$: then ind $y=\frac{1}{2}\phi(p_i^{a_i})$ for any $1\leqslant i\leqslant r$; hence there are solutions provided that $(k, \phi(p_i^{a_i}))\mid\frac{1}{2}\phi(p_i^{a_i})$, that is when $v_2(k)<\nu_i$ for any $1\leqslant i\leqslant r$; this is equivalent to $v_2(k)<\nu=\inf_i\{v_2(p_i-1)\}$.

The next lemma is useful for connecting the dyadic valuations of $n-1$ and (p_i-1).

Lemma 2. *Let $n=p_1^{a_1}\cdots p_r^{a_r}$ be an odd number, $\nu_0=v_2(n-1)$ and $\nu=\inf_{1\leqslant i\leqslant r}\{v_2(p_i-1)\}$. We always have $\nu_0\geqslant\nu$, and equality holds if and only if the number of prime factors p_i of odd exponent such that $v_2(p_i-1)=\nu$, is odd.*

Proof. Not distinguishing the prime factors of n, we write $n=q_1\cdots q_s$ where the q_i's are prime and arranged in order of ascending valuation $\nu_i=v_2(q_1-1)$; the first valuations are minimal, i.e. $\nu_1=\nu$ for $1\leqslant i\leqslant k$, and the others verify $\nu_i>\nu$, for $k<i\leqslant s$. Given $n=\prod_{1\leqslant i\leqslant s}q_i$, we write

$$n-1=\prod_{1\leqslant i\leqslant q}(q_i-1+1)-1=\sum_{1\leqslant j\leqslant q}S_j,$$

where S_j is the jth symmetric function of the (q_i-1)'s. Now we can see that S_1 contribution to $v_2(n-1)$ is preponderant; indeed, $v_2(S_j)\geqslant\nu j$ and for $j>1$, we find that $v_2(S_j)>\nu$. On the other hand,

$$v_2(S_1)=v_2\Big(k2^{\nu}+\sum_{k<i\leqslant s}(q_i-1)\Big)$$

and the second term $\sum_{k<i\leqslant s}(q_i-1)$ having a valuation $>\nu$, it follows that $v_2(S_1)\geqslant\nu$. We obtain as a first result that $\nu\leqslant\nu_0$ since $\nu_0\geqslant\inf\{v_2(S_i)\}\geqslant v_2(S_1)\geqslant\nu$; the equality is only possible if $v_2(S_1)=\nu$, and therefore also if k is odd. This concludes the proof since k has the same parity as the number of prime factors stated above.

3. Fermat's little theorem and Miller–Rabin's test

Starting from Fermat's theorem:

Theorem 1 (Fermat). *If n is a prime and a is relatively prime to n, then $a^{n-1}\equiv 1\,[n]$.*

This theorem does not characterize the prime numbers, and so it cannot be used as a reducibility test. So let K_n be the set of solutions to $a^{n-1}\equiv 1[n]$; by Lemma 1 we have $|K_n|=\prod_{1\leqslant i\leqslant r}(n-1, p_i-1)$ and since $\phi(n)=\prod_{1\leqslant i\leqslant r}p_i^{a_i-1}(p_i-1)$, we can see that $K_n=\mathbb{Z}_n^*$ only if $a_i=1$ and $(p_i-1)\mid(n-1)$ for any i, then $\lambda(n)\mid(n-1)$. Such composite numbers satisfy Fermat's test; they are called Carmichaël numbers [2] and we have proved the next result.

Theorem 2 (Carmichaël). *Both propositions are equivalent*:

(i) $\lambda(n) \mid n-1$

(ii) $a^{n-1} \equiv 1[n]$ *for any a relatively prime to n.*

The least Carmichaël number is $561 = 3 \times 11 \times 17$. It can be easily proved that Carmichaël numbers have at least three prime factors, and are square free; whether or not there are infinitely many such numbers is an open question.

Since Fermat's theorem does not provide a characterization of primes, Miller [6] has proposed to use the following refinement, which will be proved later:

Theorem 3 (Miller). *Let* $n-1 = 2^{\nu_0} n'$ *with n odd,* $d_k = (a^{n'2^k} - 1, n)$ *for* $k \in \mathbb{N}$, *and*

$$T_{\mathrm{MR}}(n,a) = \begin{cases} a^{n-1} \not\equiv 1[n] \\ \textit{or} \\ \textit{there exists an integer } 0 \leqslant k < \nu_0 \textit{ such that } d_k \neq 1 \textit{ and} \\ d_k \neq n. \end{cases}$$

Then n is composite iff $T_{\mathrm{MR}}(n, a)$ *holds for some a with* $1 \leqslant a < n$.

To convert this condition into an easily programmable and efficient algorithm, we give the equivalent formulation:

Theorem 4. *Let* $n-1 = 2^{\nu_0} n'$ *where* n' *is odd*; *the test*

$$T'_{\mathrm{MR}}(n, a) = \begin{cases} a^{n'} \not\equiv 1[n] \\ \text{and} \\ a^{n'2^k} \not\equiv -1\ [n] \textit{ for any } 0 \leqslant k < \nu_0 \end{cases}$$

is equivalent to Miller–Rabin's test.

Proof. We consider the sequence $(d_k)_{0 \leqslant k \leqslant \nu_0}$ when $T_{\mathrm{MR}}(n, a)$ is false; since $a^{n-1} \equiv 1[n]$, we find that $d_k = n$ for $k = \nu_0$. On the other hand $d_k = n$ is equivalent to $a^{n'2^k} \equiv 1[p_i^{a_i}]$, for any $1 \leqslant i \leqslant r$; since this last property stays true by squaring, it follows that $d_k = n$ implies $d_{k+1} = n$. Thus the sequence (d_k) is necessarily of the form:

(1) $d_k = n$ for any $0 \leqslant k \leqslant \nu_0$ or

(2) $d_k = 1$ for $0 \leqslant k \leqslant k_0$ and $d_k = n$ for $k_0 < k \leqslant \nu_0$.

The first case is equivalent to $a^{n'} \equiv 1[n]$. In the second we note that $x_0 = a^{n'2^{k_0}} \equiv -1[n]$; indeed, since $d_{k_0+1} = n$, we have $x_0^2 \equiv 1[p_i^{a_i}]$ for $1 \leqslant i \leqslant r$, but $x_0 - 1$ being prime to n, the only solution is $x_0 \equiv -1[n]$. It suffices to remark that the sequence $(d_k)_{0 \leqslant k \leqslant \nu_0}$ is completely defined by these conditions, in order to complete the proof.

This test allows us to build Algorithm 1 which determines whether an odd integer n is prime or not.

Algorithm 1. Repeat m times the Steps (1), (2) and (3) unless $n = 2^{\nu_0}n' + 1$ is found to be composite.

(1) Pick up a random number a, with $1 \leq a < n$.

(2) Compute $x = a^{n'} \bmod n$. If $x = 1$, then return to Step (1), otherwise go on.

(3) Squaring from x compute the sequence $a^{n'2^k} \bmod n$, where $0 \leq k < \nu_0$. If any element of this sequence equals -1, then go to Step (1) otherwise *n is composite.*

(4) The number *n is declared prime* with an error probability α_n^m.

The cost and the failure probability of this procedure will be analysed, and we shall prove in detail:

Theorem 5. *Algorithm* 1 *tests the primality of any odd number n in time* $O(m \log n\, M(n))$; *the probability of giving a composite number for a prime one is* $\alpha_n^m \leq 1/4^m$.

In the rest of this section, we prove Theorem 5 (and thus Theorem 4 and 3 which have been shown equivalent), by computing the size of the set

$$L_{\mathrm{MR}}(n) = \{1 \leq a < n \mid a^{n'} \equiv 1[n] \text{ or } a^{n'2^k} \equiv -1[n] \text{ for some } 0 \leq k < \nu_0\}.$$

When n is composite, we call *liar* an element of $L_{\mathrm{MR}}(n)$.

Size of $L_{\mathrm{MR}}(n)$: By Lemma 1, we know that there are $\prod_{1 \leq i \leq r}(n', p_i - 1)$ solutions to the equation $a^{n'} \equiv 1[n]$. Expanding on the discussion which follows this lemma, we see that $a^{n'2^k} \equiv -1[n]$ has a solution iff $k < \nu$, in which case it has $\prod_{1 \leq i \leq r}(n'2^k, p_i - 1)$ solutions. These sets of solutions being disjoint, and since $(n'2^k, p_i - 1) = 2^k(n', p_i')$, we obtain after summation:

Proposition 1. *The set* $L_{\mathrm{MR}}(n)$ *of liars to Miller–Rabin's test has size*

$$|L_{\mathrm{MR}}(n)| = \left[1 + \frac{2^{r\nu} - 1}{2^r - 1}\right] \prod_{1 \leq i \leq r} (n', p_i')$$

for n composite.

The error probability of the test T'_{MR} is $\alpha_n = |L_{\mathrm{MR}}(n)|/(n-1)$; it is useful to rewrite α_n as

$$\alpha_n = \left[1 + \frac{2^{r\nu} - 1}{2^r - 1}\right] \frac{\phi(n)}{n-1} \prod_{1 \leq i \leq r} \left(\frac{(n', p_i')}{p_i'} \frac{1}{2^{\nu_i} p_{p_i}^{a_i - 1}}\right).$$

Error Probability α_n: As an easy check, we obtain $|L_{\mathrm{MR}}(n)| = n - 1$ for n prime. For n composite, we distinguish three cases: n is a prime power, a Carmichaël number or something else.

Case 1: $n = p^a$ is a prime power. Then

$$\alpha_n = 2\frac{(n', p')}{n-1} = \frac{1}{2^{\nu-1}} \frac{1}{p^{a-1} + \cdots + p + 1} \leq \frac{1}{p+1},$$

and since $p \geq 3$ it follows that $\alpha_n \leq \frac{1}{4}$; equality $\alpha_n = \frac{1}{4}$ holds iff $n = 9$.

Case 2: n is a Carmichaël number; then n has at least three different prime factors, and each factor p_i satisfies $\alpha_i = 1$ and $p_i^r \mid n$. Our formula yields

$$\alpha_n = \left[1 - \left(1 - \frac{1}{2^{r\nu}}\right)\left(1 - \frac{1}{2^r - 1}\right)\right] \prod_{1 \leq i \leq r} 2^{\nu - \nu_i} \frac{\phi(n)}{n-1},$$

and the expression between brackets is in a decreasing function with r and ν; since $r \geq 3$ and $\nu \geq 1$, one finds that $\alpha_n \leq \frac{1}{4}\phi(n)/(n-1) < \frac{1}{4}$. The limit $\frac{1}{4}$ for α_n is approached by Carmichaël's numbers with exactly three prime factors, all of them being $\equiv 3[4]$; these numbers verify $\alpha_n = \frac{1}{4}\phi(n)/(n-1)$. Examples are: $n = 8911$, with $\alpha_n = \frac{1}{5}$, and $n = 1024651$ with $\alpha_n = 27/(5 \times 23) \simeq 0.23478$.

Case 3: n has at least two prime factors but it is not a Carmichaël number. So there exists a factor p_j such that $p_j - 1$ does not divide $n - 1$; therefore $\nu_j > \nu_0$ or n' is not a multiple of p_i'. In the first case, we note that $\Sigma_{1 \leq i \leq r}(\nu_i - \nu)$ is ≥ 1, and because $r \geq 2$, it follows that $\alpha_n < \frac{1}{4}$. In the other case, since the odd number p_j' does not divide n', this implies that $(n', p_j')/p_j' \leq \frac{1}{3}$ and so $\alpha_n < \frac{1}{6}$. Numbers having exactly two prime factors that verify $p_i' \mid n'$ are approaching the limit $\frac{1}{4}$; such numbers are on the form $n = (2p+1)(4p+1)$, where $(2p+1)$ and $(4p+1)$ are prime and p is odd. Examples are $n = 15$, $n = 91$; $n = 703$, $n = 1891$ and a larger one is $n = 497503$; if the set is infinite (open question), then α_n can be arbitrarily close to $\frac{1}{4}$; indeed when $n = (2p+1)(4p+1)$, one finds that

$$\alpha_n = \frac{1}{4}\left(1 - \frac{1}{4p+3}\right).$$

We have proved the following proposition:

Proposition 2. *The failure probability α_n of Miller–Rabin's test is less than $\frac{1}{4}$ for any composite n.*

One remarks that we cannot presently give an upper bound lower than $\frac{1}{4}$ since we do not know whether the set of numbers approaching this value is finite or not.

We end the analysis of Algorithm 1 by computing its cost; one test requires the computation of $a^{(n-1)/2} \bmod n$, whose cost is $O(\log n\, M(n))$ [5] and we compute at most m such tests: the cost of whole Algorithm 1 is $O(m \log n\, M(n))$.

4. Euler's criterion and Solovay–Strassen test

This section starts with Euler's criterion:

Theorem 6 (Euler). *Let n be an odd prime. Then any a prime to n satisfies $\left(\frac{a}{n}\right) a^{(n-1)/2} \equiv 1[n]$.*

Solovay and Strassen [9] have shown a converse of this:

Theorem 7 (Solovay–Strassen). *The number n is prime iff $(\frac{a}{n})\, a^{(n-1)/2} \equiv 1[n]$ for all $1 \leq a < n$.*

Solovay and Strassen prove that the relation $T_{ss}(n, a) = (\frac{a}{n}) a^{(n-1)/2} \not\equiv 1[n]$ is a primality test in the sense defined in the introduction of this paper. They infer Algorithm 2 which tests whether n is prime or not:

Algorithm 2. Repeat m times the Steps (1) and (2), unless n is declared composite.
(1) take a random number a from a uniform distribution on $1 \leq a < n$.
(2) If $(\frac{a}{n}) a^{(n-1)/2} \not\equiv 1[n]$ then *n is composite* otherwise go back to (1).
(3) *n is prime* with an error probability β_n^m.

Computing $(\frac{a}{n})$ is isomorphic to computing (a, n); this can be done [5] in $O(\log^2 n)$, and thus the cost of whole algorithm is of complexity $O(\log n\, M(n))$.

We analyse the error probability of this algorithm and prove:

Theorem 8 (Solovay–Strassen). *Let $T_{ss}(n, a)$ be the test $(\frac{a}{n}) a^{(n-1)/2} \not\equiv 1[n]$ definite for any odd n and any $a \in \mathbb{Z}_n^*$.*

Then the set $L_{ss}(n)$ of liars associated with T_{ss} has at most

$$\frac{\phi(n)}{2} \quad \textit{elements.}$$

To prove this result, we introduce the following characters on $\mathbb{Z}_n^*$ and their kernels:

$$E : a \to a^{(n-1)/2} \bmod n, \qquad D_n = \ker E = \{a \in \mathbb{Z}_n^*;\ a^{(n-1)/2} \equiv 1[n]\},$$

$$J : a \to (\tfrac{a}{n})$$

$$V : a \to E(a) \times J(a), \qquad L_{ss}(n) = \ker V = \{a \in \mathbb{Z}_n^*;\ a^{(n-1)/2} (\tfrac{a}{n}) \equiv 1[n]\}.$$

The error probability of T_{ss} is $\beta_n = |L_{ss}(n)|/(n-1)$, so we have to compute the size of $L_{ss}(n)$.

Size of $L_{ss}(n)$: We note $|L_{ss}(n)| = \delta_n |D_n|$.

Lemma 1 gives $|D_n| = \prod_{1 \leq i \leq r} (\frac{1}{2}(n-1), p_i - 1)$. Since J takes on $\mathbb{Z}_n^*$ at most the values 1 and -1, it follows that $\delta_n = 1, \frac{1}{2}$ or 2. The exact value of δ_n will be discussed according to whether the characters E or V take the value -1 or not.

Calling g a generator of $\mathbb{Z}_{p^\alpha}^*$, where p is any prime factor of $n = p_1^{a_1} \cdots p_r^{a_r}$, we derive as consequences of Lemma 1 and 2:

If $E(a) = 1$, then the index of a relative to g is

$$\lambda = t \frac{p^{\alpha-1}(p-1)}{(\frac{1}{2}(n-1), p-1)}, \quad \text{where } t \in \mathbb{Z}$$

and if $E(a)=-1$, then $\mathrm{ind}_g a$ is

$$\lambda=(2t+1)\frac{p^{\alpha-1}(p-1)}{(n-1,p-1)} \quad \text{with } t\in\mathbb{Z};$$

the last case occurs only if $\nu=\nu_0$.

Furthermore, g being generator is not a square in $\mathbb{Z}_{p^\alpha}$, and so $(\frac{g}{p})=-1$; coming back to n we derive by multiplicative properties of J

$$J(a)=(-1)^{\sum_{1\le i\le r}(a_i\lambda_i)}.$$

Now let us distinguish the two cases:

Case 1: $\mathrm{Im}\,E$ contains the value -1; then one knows that $\nu=\nu_0$. If $E(a)=1$, then $v_2(\lambda_i)\geqslant \nu_i+1-\nu_0\geqslant 1$ and all indexes of a are even.

Hence, $J(a)=1$ and so $V(a)=1$. On the contrary, if $E(a)=-1$, we find that $v_2(\lambda_i)=\nu_i-\nu$ and Lemma 3 states that there is an even number of prime factors p_i such as a_i is odd and $\nu_i=\nu$; consequently $\sum_{1\le i\le r}a_i\lambda_i$ is even and $J(a)=-1$.

Again we obtain $V(a)=1$. Thus -1 cannot belong both to $\mathrm{Im}\,E$ and $\mathrm{Im}\,V$; we deduce from this $\delta_n=|\mathrm{Im}\,E|/|\mathrm{Im}\,V|=2$.

Case 2: -1 is not in $\mathrm{Im}\,E$; $V(a)=-1$ implies that $E(a)=1$ and $J(a)=-1$.

On condition that there exists an odd $(a_j\lambda_j)$ and choosing for example $t_j=1$ and $t_i=2$ for $i\neq j$, we build up an integer a such as $V(a)=-1$; since $\nu<\nu_0$, index λ_i is odd when $\nu_i<\nu_0$, and it follows that: if $\nu_i<\nu_0$ for some i with a_i odd, then $\delta_n=\frac{1}{2}$, else $\delta_n=1$.

We sum up with:

Proposition 3. *The set of a's which are liars to Solovay–Strassen's test is of size*

$$|L_{ss}(n)|=\delta_n\prod_{1\le i\le r}(\tfrac{1}{2}(n-1),p_i-1),$$

$$\text{where } \delta_n=\begin{cases}2, & \text{if } \nu=\nu_0,\\ \frac{1}{2}, & \text{if } \nu_i<\nu_0 \text{ for some } i \text{ with } a_i \text{ odd},\\ 1, & \text{otherwise.}\end{cases}$$

We can now prove the Theorem 8; let us consider

$$\beta'_n=\frac{|L_{ss}(n)|}{n-1}=\delta_n 2^{-\sigma}\prod_{1\le i\le r}\frac{(n',p'_i)}{p^{\alpha_i-1}p'_i},$$

where $\sigma=\sum_{1\le i\le r}\max\{0,\nu_i-\nu_0+1\}$.

The error probability β_n of Solovay–Strassen's is $\beta_n=\beta'_n\phi(n)/(n-1)<\beta'_n$, thus we only need to bound β'_n.

We distinguish two cases, according to whether $\nu=\nu_0$ or $\nu<\nu_0$.

$\nu=\nu_0$. We know that $\delta_n=2$. If n is a prime power, we find that $\beta'_n\leqslant\frac{1}{3}$; otherwise we divide the discussion as follows.

- $r=2$, and by Lemma 2, it follows that $\sigma \geqslant 3$, from which we have $\beta'_n \leqslant \frac{1}{4}$.
- $r \geqslant 3$, and since $\sigma \geqslant r$ we find again that $\beta'_n < \frac{1}{4}$.

$\nu < \nu_0$. Here $\delta_n = 1$ or $\delta_n = \frac{1}{2}$. If there exists some $a_i > 1$, then n is not square free and we find $\beta'_n \leqslant 1/p_i \leqslant \frac{1}{3}$; else we have $\delta_n = \frac{1}{2}$ because every a_i is odd and because there exists $\nu_i < \nu_0$; this yields $\beta'_n \leqslant \frac{1}{2}$.

The numbers achieving the bound $\beta'_n = \frac{1}{2}$ verify

$$p'_i \mid n, \quad \nu_i < \nu_0 \text{ and } a_i = 1 \qquad \text{for any } 1 \leqslant i \leqslant r.$$

They are a subset of Carmichaël numbers, and examples are $n = 1729$[1], $n = 2465$, $n = 15841$, $n = 41041$, $n = 46657$, $n = 75361$, $n = 162401$.

Thus for any composite n, we have found that $\beta_n < \frac{1}{2}$, thus completing the proof of Theorem 8; (if n is prime, then $\beta'_n = \beta_n = 1$, and this also provides a proof of Euler's criterion).

As in the first section, we cannot improve on the $\frac{1}{2}$ bound for β_n, since we do not know whether or not there are infinitely many Carmichaël numbers of the above type.

To conclude this section, we can state:

Algorithm 2 *tests in time* $O(m \log n\, M(n))$ *whether or not an odd integer n is prime, with a failure probability* $<1/2^m$.

5. Comparison of the two tests

It remains to be proved that Miller–Rabin's test is always better than Solovay–Strassen's. Both tests basically compute $a^{(n-1)/2} \bmod n$, which gives an asymptotic cost $O(\log n\, M(n))$, but Solovay–Strassen's requires to compute $\left(\frac{a}{n}\right)$ in addition.

Furthermore we prove inclusion between the tests, in that any liar to Miller–Rabin's test is also a liar to Solovay–Strassen's.

Theorem 9. *If an integer a does not verify Miller–Rabin's test* $T_{\mathrm{MR}}(n, a)$ *neither does it verify Solovay–Strassen's.*

Proof. We use the notations of Section 4. An integer a which does not verify $T_{\mathrm{MR}}(n, a)$ is in $L_{\mathrm{MR}}(n)$; let us prove that it is also an element of $L_{\mathrm{ss}}(n)$, by distinguishing three cases.

Case 1: $a^{n'} \equiv 1[n]$. By squaring, we find that $E(a) = 1$; on the other hand $\lambda_i = t_i \phi(p_i^{a_i})/(n', p'_i)$ is even because (n', p'_i) is odd and $\phi(p_i^{a_i})$ is even; it follows that $J(a) = 1$, from which one concludes that $V(a) = 1$, and so $a \in L_{\mathrm{ss}}(n)$.

Case 2: $a^{(n-1)/2} \equiv -1[n]$, so $E(a) = -1$ and one knows (see Section 4) that necessarily $J(a) = -1$; again we find that $a \in L_{\mathrm{ss}}(n)$.

[1] Knuth remarks that 1729 is the smallest Ramanujan number, i.e. it has two representations as a sum of cubes: $1729 = 10^3 + 9^3 = 12^3 + 1^3$.

Case 3. $a^{n'2^k} \equiv -1[n]$ for some k with $0 \leq k < \nu$. The index $\lambda_i = \phi(p_i^{a_i})/(n'2^{k+1}, p_i - 1)$ is odd iff $\nu_i \leq k+1$, thus $\nu_i = \nu = k+1$; from this we deduce that $\nu = k+1 < \nu_0$, and by Lemma 1 it follows that $\sum_{1 \leq i \leq r} (a_i \lambda_i)$ is even; finally we have $J(a) = 1$ and $a \in L_{ss}(n)$.

We can refine this result by studying the ratio

$$\rho_n = \frac{|L_{MR}(n)|}{|L_{ss}(n)|} = \frac{\alpha_n}{\beta_n}.$$

From expressions of α_n and β_n we derive

$$\rho_n = \frac{1}{\delta_n} 2^{-\sigma} \left[1 + \frac{2^{r\nu} - 1}{2^r - 1} \right] \quad \text{with } \sigma = \textstyle\sum_{1 \leq i \leq r} \inf\{\nu_0 - 1, \nu_i\},$$

and we discuss the value of ρ_n according to $\nu = \nu_0$ or $\nu < \nu_0$.

Case 1: $\nu = \nu_0$. From the proof of Theorem 8, we are in the case $\delta_n = 2$ and $\sigma = r(\nu - 1)$, from which we find that

$$\rho_n = 2^{r-1} \left[1 - \left(1 - \frac{1}{2^r - 1}\right)\left(1 - \frac{1}{2^{r\nu}}\right) \right].$$

This expression decreases with r and ν, and its maximum $\rho_n = 1$ is reached if $r = 1$ or if $\nu = 1$; for any other value of r and ν we find that

$$\frac{1}{2}\left(1 + \frac{1}{2^r - 1}\right) < \rho_n \leq \frac{1}{2}\left(1 + \frac{1}{2^{r-1}}\right).$$

Case 2: $\nu < \nu_0$. If $r = 1$, then n is a prime power and $\delta_n = 1$, from which we find that $\rho_n = 1$; otherwise we rewrite

$$\rho_n \leq 2\left[1 - \left(1 - \frac{1}{2^{r\nu}}\right)\left(1 - \frac{1}{2^r - 1}\right) \right],$$

and the maximum $\rho_n = 1$ is reached by the numbers $n = p_1^{a_1} p_2^{a_2}$ with $\nu_1 = \nu_2 = 1$ and both exponents odd; for any other number we have $\rho_n < 2^{2-r} < 1$.

The ratio ρ_n can take the value 1, in which case both tests are equivalent; it can also be arbitrarily small, for example if n verifies $1 < \nu < \nu_0$ and $r > 2$ in which case $\rho_n < 2^{2-r}$. In order to study the average value of ρ_n, we introduce

$$\tilde{\rho}_n = \frac{1}{|A_n|} \sum_{m \in A_n} \rho_m, \quad \text{where } A_n = \{m \text{ odd}, m \leq n\}.$$

We prove that $\tilde{\rho}_n$ has a limit $\frac{1}{2}$ as n goes to infinity. This is achieved by decomposing A_n in four sets; we feel free to use standard results from [3] as we need them.

Set 1 is the set of prime numbers $\leq n$; let us agree that $\rho_n = 1$, and we find that $\sum_{\text{Set 1}} \rho_m = O(n/\log n)$.

Set 2 contains the composite numbers which verify $\rho_m = 1$; one can find

- integers $m \equiv 3[4]$, in number $\frac{1}{4}n + O(n/\log n)$,

- prime powers $m = p^a$, in number $O(n/\log n)$,
- integers $m = p_1^{a_1} p_2^{a_2}$ as above, of which there are $O((n \operatorname{llog} n)/\log n)$; thus it follows that

$$\sum_{\text{Set}2} \rho_m = \tfrac{1}{4}n + O\left(\frac{n \operatorname{llog} n}{\log n}\right).$$

Set 3: its elements verify $\nu = \nu_0 > 1$ and $r > 2$; we over-estimate it by counting the numbers m with $\nu > 1$. Let $P_3(p)$ be the property

$$p \text{ is prime}, \qquad r \equiv 3[4] \quad \text{and} \quad p \leq n.$$

We find that the number of integers m with $\nu > 1$ is

$$\sum_{P_3(p)} \left\lfloor \frac{n}{p} \right\rfloor - \sum_{\substack{P_3(p) \\ P_3(q)}} \left\lfloor \frac{n}{pq} \right\rfloor + \cdots \leq n \prod_{P_3(p)} \left(1 - \frac{1}{p}\right),$$

and the product $\prod_{P_3(p)} (1 - 1/p)$ is computed in [4] and its value is $O(n/\sqrt{\log n})$; it follows that

$$\sum_{\text{Set}3} \rho_m = O\left(\frac{n}{\sqrt{\log n}}\right).$$

Set *4:* all the others composite numbers $m \leq n$ are in Set 4. They verify $\nu < \nu_0$ and $r \geq 2$; because $\rho_n \leq 2^{2-r}$ it follows that

$$\sum_{\text{Set}4} \rho_m \leq \sum_{\text{Set}4} 2^{2-r} \leq K \sum_{r \geq 2} \frac{(\operatorname{llog} n)^{r-1}}{(r-1)!} 2^{2-r}$$

$$< \frac{4Kn}{\log n} e^{(\operatorname{llog} n)/2}$$

and so

$$\sum_{\text{Set}4} \rho_m = O\left(\frac{n}{\sqrt{\log n}}\right).$$

Summation yields

$$\tfrac{1}{4}n + O\left(\frac{n \operatorname{llog} n}{\log n}\right) < \sum_{m \in A_n} \rho_m < \tfrac{1}{4}n + O\left(\frac{n}{\sqrt{\log n}}\right),$$

and because $|A_n| = \frac{1}{2}n$, we find that $\tilde{\rho}_n = \frac{1}{2} + O(1/\sqrt{\log n})$, thus:

Proposition 4. *The average value of the ratio ρ_n between error probabilities of the two primality tests has limit $\frac{1}{2}$ when $n \to \infty$.*

This finishes the analysis and the comparison of tests; we sum up in the conclusion.

6. Conclusion

Miller–Rabin's test is used in an algorithm which determines whether an odd number is prime or not in time $O(m \log n\, M(n))$; the probability that it erroneously calls 'prime' a composite number is less than 4^{-m}.

Solovay–Strassen's is only a subtest of the first test; it is always more expansive and on an average its failure probability is twice that of Miller–Rabin's.

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